Qiyaf

— Elevating Tomorrow —

Vision-Based Object Detection for UAV Solar Panel Inspection Using Deep Learning

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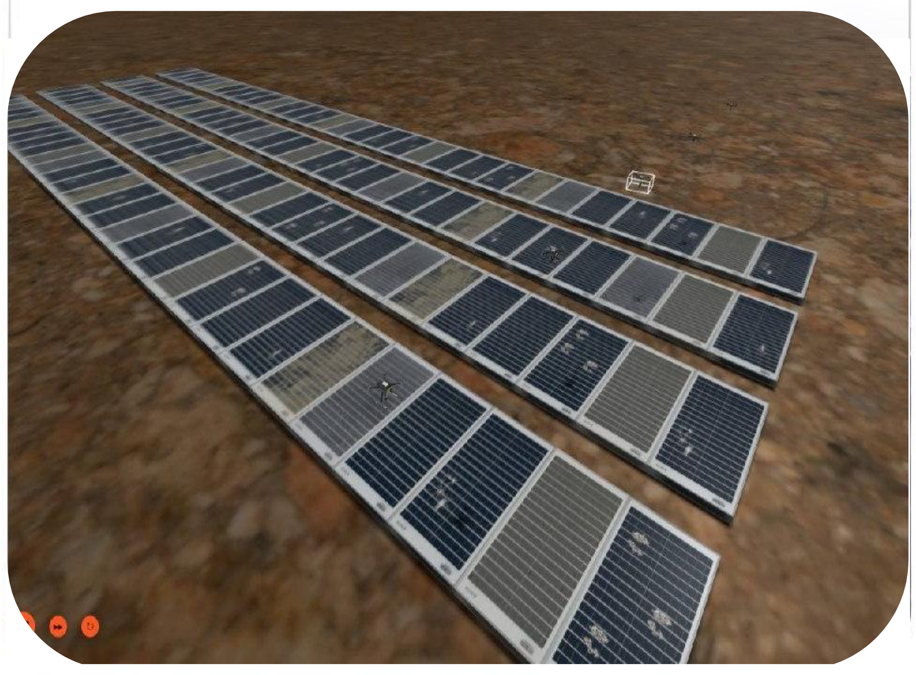
INTRODUCTION

Solar energy plays a critical role in sustainable power generation. Manual inspection methods are time-consuming and error-prone. Our system uses UAVs and deep learning to automatically detect defects such as cracks, snow, dust, and bird droppings, enabling accurate assessment and efficient maintenance.



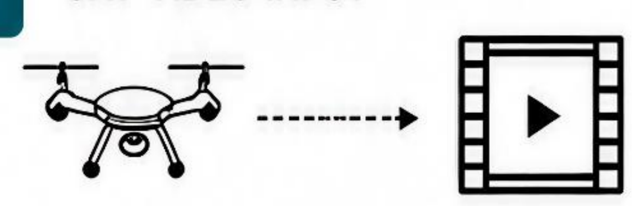
OBJECTIVES

- Automate large-scale solar panel inspection using UAVs
- Develop **a two-stage AI system** for panel and defect detection
- Reduce reliance on manual inspection methods
- Enhance detection stability using tracking and voting
- Achieve near- real-time analysis through chunk-based processing
- Deliver a scalable and cost-effective inspection solution

HOW DO THE MODELS WORK ?

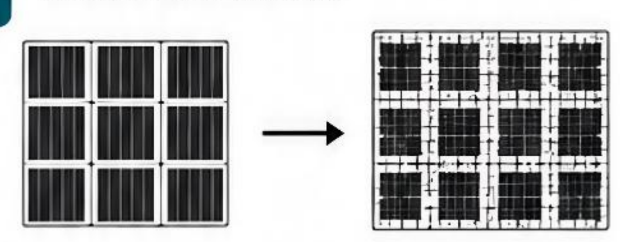
PIPELINE MODELS

1 UAV VIDEO INPUT



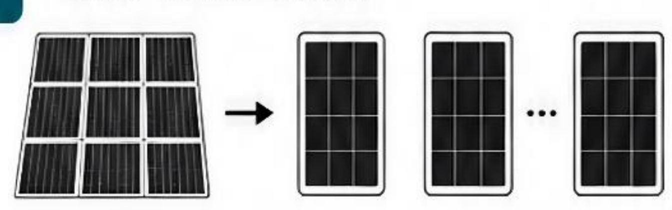
The system receives video input captured by the UAV in 10-second chunks for efficient processing.

2 PANEL DETECTION



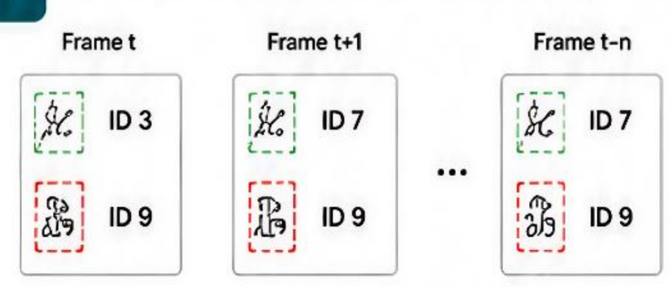
Solar panels are detected using the YOLOv11 frame model with Panel Detection Model.

3 CROP EXTRACTION



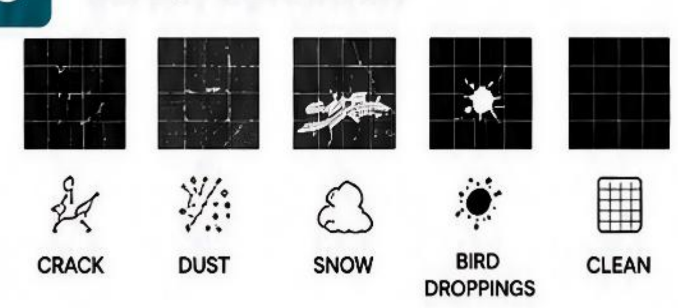
Each detected panel is automatically cropped into a focused image region for detailed analysis.

4 PANEL TRACKING (BYTETRACK)



The cropped panel is passed to ByteTrack, where each panel is assigned a unique ID for consistent tracking across frames.

5 DEFECT DETECTION



Each tracked panel is analyzed using the Defect Detection Model to identify defects such as: **Cracks**,Dust, Snow, Bird Droppings, or Clean Panels.

6 VOTING SYSTEM

Detection Scores

Frame 1	Frame 2	Frame3
Panel 1	CRACK	DUST
Panel 2	G0ANT	DuST
...	...	...
Panel n	CLEAN	0.034

→ Hybrid Voting Mechanism → Final Decision: CRACK 0.87

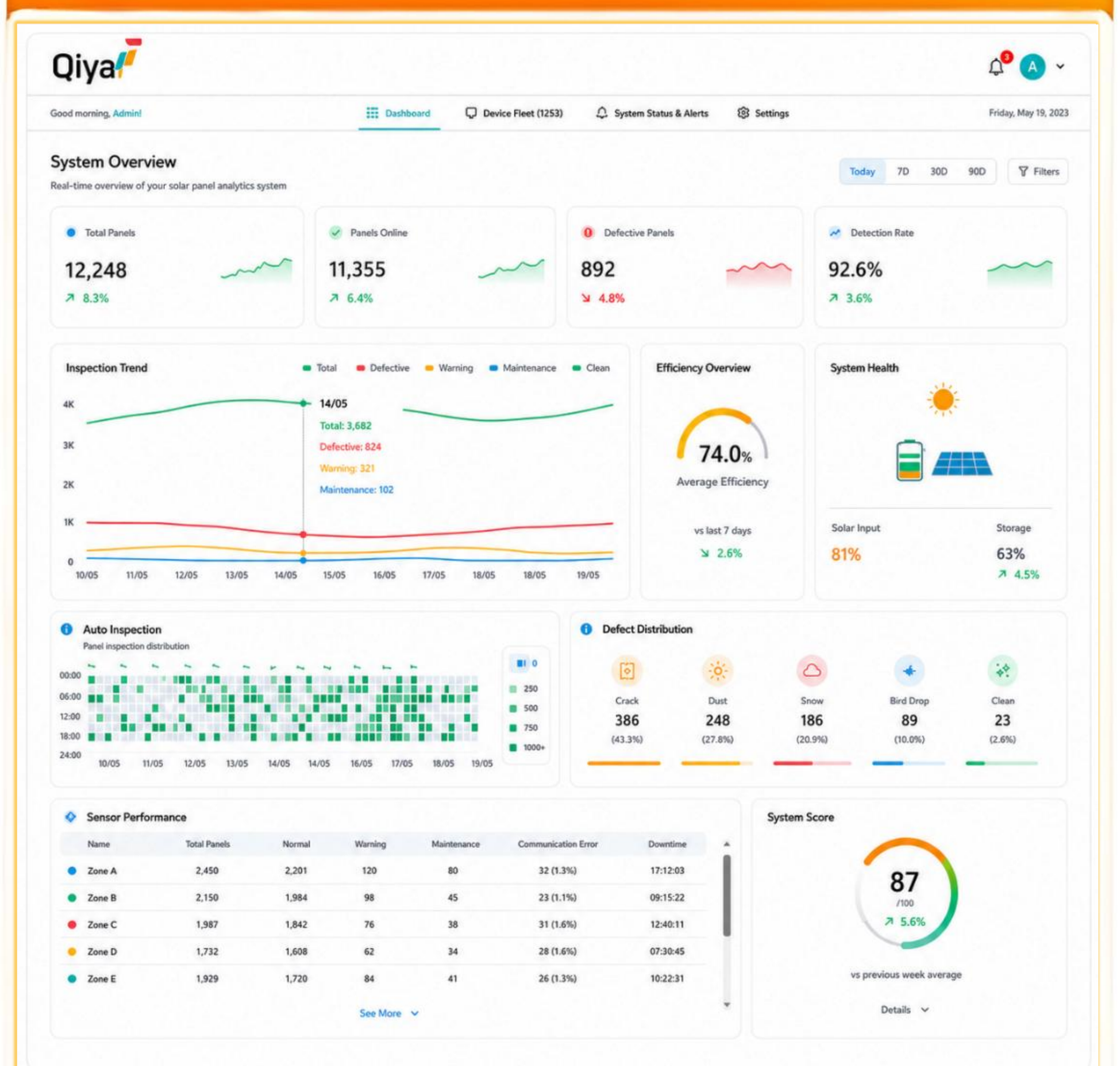
A Hybrid Voting mechanism combines repeated predictions with confidence scores to determine the most reliable final label.

7 FINAL OUTPUT

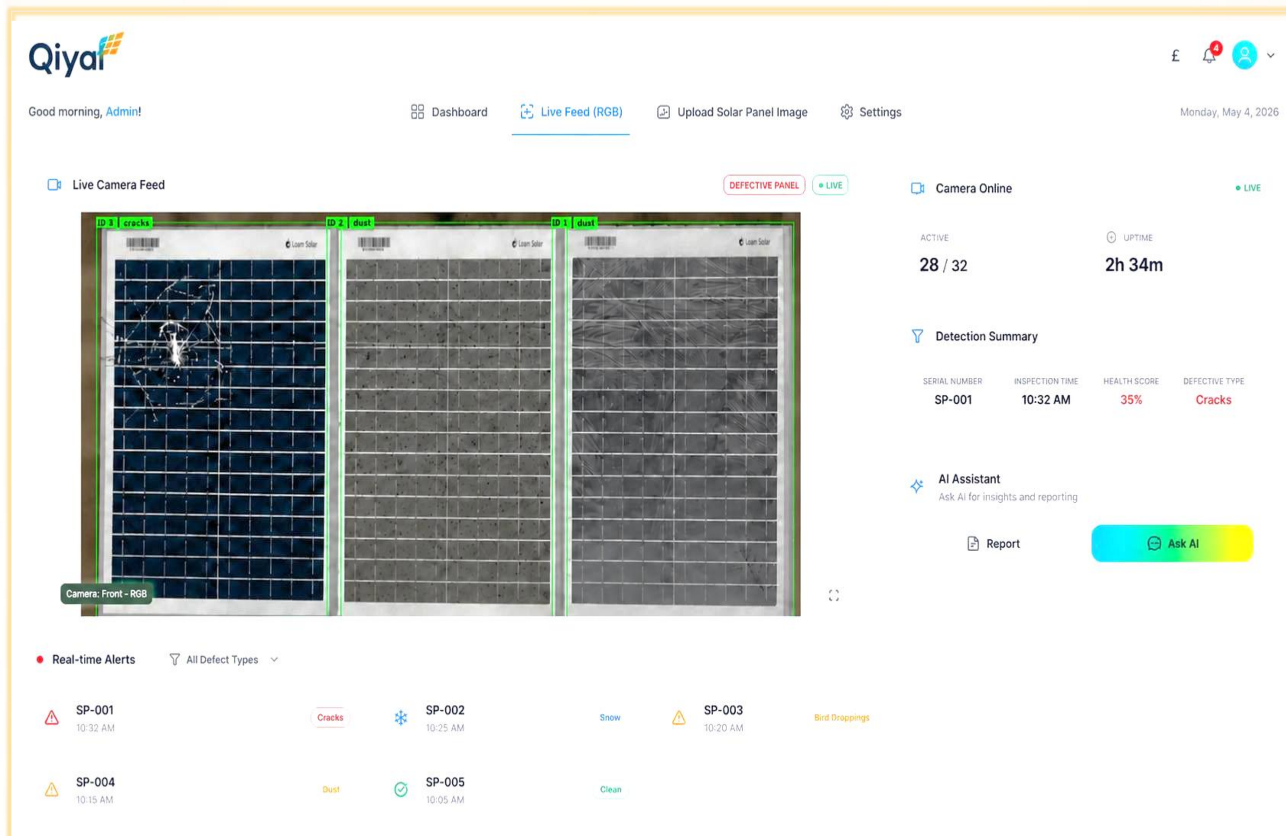
ID	Defect Label	Confidence	Timestamp
ID 3	CRACK (0.96)	0.96	2025-05-19 10:15:23
ID 7	DUST (0.89)	0.89	2025-05-19 10:15:34
ID 9	CLEAN (0.98)	0.98	2025-05-19 10:15:35
...	...	...	...

The system produces an accurate defect result linked to each tracked panel ID, enabling reliable monitoring and reporting.

RESULTS



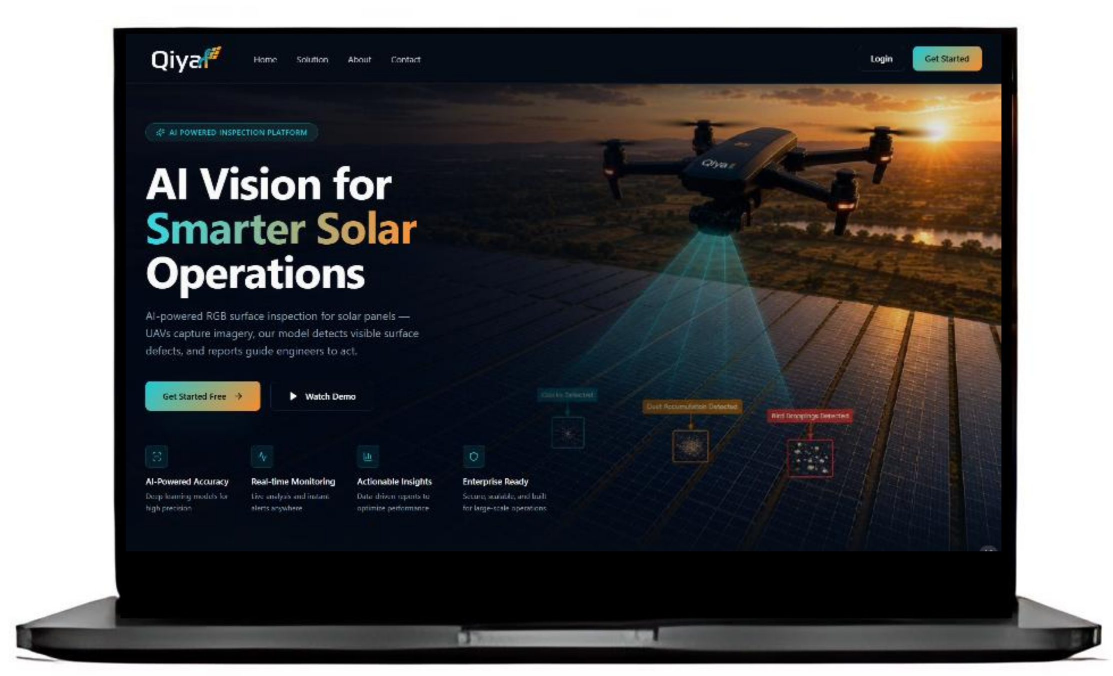
Dashboard Interface: Displays solar panel data collected by the UAV including system status,defect statistics ,performance analysis



Live Feed Interface: Displays the UAV's near-real-time camera feed, highlights detected solar panel defects, and provides live status updates, detection summaries, and real-time alerts.

TOOLS

ultralytics YOLO · GitHub · docker · React.js · FastAPI



DEMO

Scan to watch demo

METHODOLOGY

1. **UAV Video Capture** — Capture video of solar panels using UAVs
2. **Video Chunk (10s segments)** — Split videos into 10-second chunks for efficient processing
3. **Frame Extraction** — Extract frames from each chunk for input to AI models
4. **AI Pipeline (Detection & Tracking)** — Detect and track defects using deep learning models for accurate identification
5. **Dashboard & Results View** — Visualize inspection reports, live analytics, and performance dashboards